

MATH 130: 9/29 WORKSHEET

Last time we talked about how truth tables could be used to give a procedure for checking validity of an argument. Now we look at using truth tables to make sense of other concepts.

Two sentences P and Q are *logically equivalent* if their truth tables have the same pattern. That is, sentences are logically equivalent if they behave identically for any possible combination of inputs.

Some of these pairs of equivalent sentences we have already seen.

- $A \Leftrightarrow B$ and $(A \Rightarrow B) \wedge (B \Rightarrow A)$
- $A \wedge B$ and $B \wedge A$
- $(A \wedge B) \wedge C$ and $A \wedge (B \wedge C)$
- $A \vee B$ and $B \vee A$
- $(A \vee B) \vee C$ and $A \vee (B \vee C)$
- $A \Rightarrow B$ and $\neg B \Rightarrow \neg A$

- Two (or more) sentences P and Q are *jointly consistent* if there is a row where both their truth tables are true. That is, sentences are jointly consistent if it's possible for them both to be true at once. If we're talking about a single sentence, we say it's consistent if it's possible for it to be true.
- A sentence P is a *contradiction* if every row in its truth table is false. That is, a contradiction is a sentence which is impossible.
- A sentence P is a *tautology* if every row in its truth table is true.
- A sentence P is *contingent* if it is true in some row and false in some other row of its truth table. That is, P is contingent if P and $\neg P$ are both consistent.

We can classify statements as contradictions (always false), contingent (sometimes false, sometimes true), or tautologies (always true).

- $A \wedge \neg A$
- $A \vee \neg A$
- $A \Rightarrow A$
- $A \Rightarrow \neg A$
- $A \Leftrightarrow \neg A$
- $(A \wedge \neg B) \vee (B \wedge \neg A)$
- $(A \Rightarrow B) \vee (B \Rightarrow A)$
- $(A \Rightarrow B) \vee (C \Rightarrow B)$
- $(A \Rightarrow B) \vee (B \Rightarrow C) \vee (C \Rightarrow A)$